

Numerical Methods Lab

Lab 19 – 26 | Python Program Outputs

Terminal Output — Python

Lab-19: Jacobi Iteration Method

Question: Write a program for Jacobi Iteration Method. $4x + y + 2z = 4$, $3x + 5y + z = 7$, $x + y + 3z = 3$

Python Output:

```
big@hell-na:~/Kishor/nm$ python3 lab19.py
Enter the coefficients for the system of equations (ax + by + cz = d):

Equation 1:
Coefficient of x (a1): 4
Coefficient of y (b1): 1
Coefficient of z (c1): 2
Constant term (d1): 4

Equation 2:
Coefficient of x (a2): 3
Coefficient of y (b2): 5
Coefficient of z (c2): 1
Constant term (d2): 7

Equation 3:
Coefficient of x (a3): 1
Coefficient of y (b3): 1
Coefficient of z (c3): 3
Constant term (d3): 3

Iter | x | y | z | Max Error
-----
1 | 1.000000 | 1.400000 | 1.000000 | 1.400000e+00
2 | 0.150000 | 0.600000 | 0.200000 | 8.500000e-01
3 | 0.750000 | 1.270000 | 0.750000 | 6.700000e-01
...
46 | 0.500000 | 0.999999 | 0.500000 | 1.225186e-06
47 | 0.500000 | 1.000000 | 0.500000 | 9.018472e-07

Converged in 47 iterations.
Final Solution: x=0.500000, y=1.000000, z=0.500000
big@hell-na:~/Kishor/nm$
```

Lab-20: Gauss-Seidel Method

Question: Write a program for Gauss-Seidel Method. $4x + y + 2z = 4$, $3x + 5y + z = 7$, $x + y + 3z = 3$

Python Output:

```
big@hell-na:~/Kishor/nm$ python3 lab20.py
Enter the coefficients for the system (ax + by + cz = d):
```

Equation 1:

```
Coefficient of x (a1): 4
Coefficient of y (b1): 1
Coefficient of z (c1): 2
Constant term (d1): 4
```

Equation 2:

```
Coefficient of x (a2): 3
Coefficient of y (b2): 5
Coefficient of z (c2): 1
Constant term (d2): 7
```

Equation 3:

```
Coefficient of x (a3): 1
Coefficient of y (b3): 1
Coefficient of z (c3): 3
Constant term (d3): 3
```

```
Iter | x | y | z | Max Error
```

```
-----
1 | 1.000000 | 0.800000 | 0.400000 | 1.000000e+00
2 | 0.600000 | 0.960000 | 0.480000 | 4.000000e-01
3 | 0.520000 | 0.992000 | 0.496000 | 8.000000e-02
4 | 0.504000 | 0.998400 | 0.499200 | 1.600000e-02
5 | 0.500800 | 0.999680 | 0.499840 | 3.200000e-03
6 | 0.500160 | 0.999936 | 0.499968 | 6.400000e-04
7 | 0.500032 | 0.999987 | 0.499994 | 1.280000e-04
8 | 0.500006 | 0.999997 | 0.499999 | 2.560000e-05
9 | 0.500001 | 0.999999 | 0.500000 | 5.120000e-06
10 | 0.500000 | 1.000000 | 0.500000 | 1.024000e-06
11 | 0.500000 | 1.000000 | 0.500000 | 2.048000e-07
```

Converged in 11 iterations.

Final Solution: x=0.500000, y=1.000000, z=0.500000

```
big@hell-na:~/Kishor/nm$
```

Lab-21: Taylor Series Method

Question: Write a program for Taylor series expansion. $dy/dx = 2y/x$, $x_0=4$, $y_0=6$, $h=0.25$, evaluate at $x=6$

Python Output:

```
big@hell-na:~/Kishor/nm$ python3 lab21.py
Taylor Series Method for dy/dx = 2y/x
-----
Enter initial x (x0): 4
Enter initial y (y0): 6
Enter target x value: 6
Enter step size (h): 0.25

Step | x | y
-----
0 | 4.0000 | 6.0000
1 | 4.2500 | 6.7734
2 | 4.5000 | 7.5938
3 | 4.7500 | 8.4609
4 | 5.0000 | 9.3750
5 | 5.2500 | 10.3359
6 | 5.5000 | 11.3438
7 | 5.7500 | 12.3984
8 | 6.0000 | 13.5000
-----
Approximate solution at x = 6.0 is y = 13.500000
big@hell-na:~/Kishor/nm$
```

Lab-22: Euler's Method

Question: Write a program for Euler's method. $dy/dx = 2y/x$, $x_0=4$, $y_0=6$, $h=0.25$, evaluate at $x=6$

Python Output:

```
big@hell-na:~/Kishor/nm$ python3 lab22.py
Euler's Method Solver for dy/dx = 2y/x
-----
Enter initial x (x0): 4
Enter initial y (y0): 6
Enter target x value to solve for: 6
Enter step size (h): 0.25

Step | x | y
-----
0 | 4.0000 | 6.0000
1 | 4.2500 | 6.7500
2 | 4.5000 | 7.5441
3 | 4.7500 | 8.3824
4 | 5.0000 | 9.2647
5 | 5.2500 | 10.1912
6 | 5.5000 | 11.1618
7 | 5.7500 | 12.1765
8 | 6.0000 | 13.2353
-----
Approximate solution at x = 6.0 is y = 13.235294
big@hell-na:~/Kishor/nm$
```

Lab-23: Heun's Method

Question: Write a program for Heun's method. $dy/dx = x + y$, $x_0=0$, $y_0=1$, $h=0.1$, evaluate at $x=1$

Python Output:

```
big@hell-na:~/Kishor/nm$ python3 lab23.py
```

```
Heun's Method Solver for  $dy/dx = x + y$ 
```

```
-----
```

```
Enter initial x (x0): 0
```

```
Enter initial y (y0): 1
```

```
Enter target x value: 1
```

```
Enter step size (h): 0.1
```

```
Step | x | y | Slope Avg
```

```
-----
```

```
0 | 0.0000 | 1.0000 | N/A
```

```
1 | 0.1000 | 1.1100 | 1.1000
```

```
2 | 0.2000 | 1.2421 | 1.3205
```

```
3 | 0.3000 | 1.3985 | 1.5642
```

```
4 | 0.4000 | 1.5818 | 1.8334
```

```
5 | 0.5000 | 1.7949 | 2.1309
```

```
6 | 0.6000 | 2.0409 | 2.4596
```

```
7 | 0.7000 | 2.3231 | 2.8229
```

```
8 | 0.8000 | 2.6456 | 3.2243
```

```
9 | 0.9000 | 3.0124 | 3.6679
```

```
10 | 1.0000 | 3.4282 | 4.1580
```

```
-----
```

```
Approximate solution at  $x = 1.0$  is  $y = 3.428162$ 
```

```
big@hell-na:~/Kishor/nm$
```

Lab-24: Runge-Kutta 4th Order (RK4) Method

Question: Write a program for RK4 method. $dy/dx = 3x + y$, $x_0=2$, $y_0=4$, $h=0.1$, evaluate at $x=3$

Python Output:

```
big@hell-na:~/Kishor/nm$ python3 lab24.py
RK4 Method Solver for dy/dx = 3x + y
-----
Enter initial x (x0): 2
Enter initial y (y0): 4
Enter target x value: 3
Enter step size (h): 0.1

Step | x | y
-----
0 | 2.0000 | 4.0000
1 | 2.1000 | 5.0672
2 | 2.2000 | 6.2782
3 | 2.3000 | 7.6482
4 | 2.4000 | 9.1937
5 | 2.5000 | 10.9334
6 | 2.6000 | 12.8875
7 | 2.7000 | 15.0788
8 | 2.8000 | 17.5320
9 | 2.9000 | 20.2748
10 | 3.0000 | 23.3376
-----
Approximate solution at x = 3.0 is y = 23.337637
big@hell-na:~/Kishor/nm$
```

Lab-25: Laplace Equation

Question: Write a program for Laplace equation using finite difference method.

Python Output:

```
big@hell-na:~/Kishor/nm$ python3 lab25.py
Enter the size of the grid (n x n): 3

0.0000 1.0000 1.0000
0.0000 0.5000 1.0000
0.0000 0.0000 1.0000
big@hell-na:~/Kishor/nm$
```


[illegible]